

Fig. 5: Block diagram of the fan speed increasing regulator

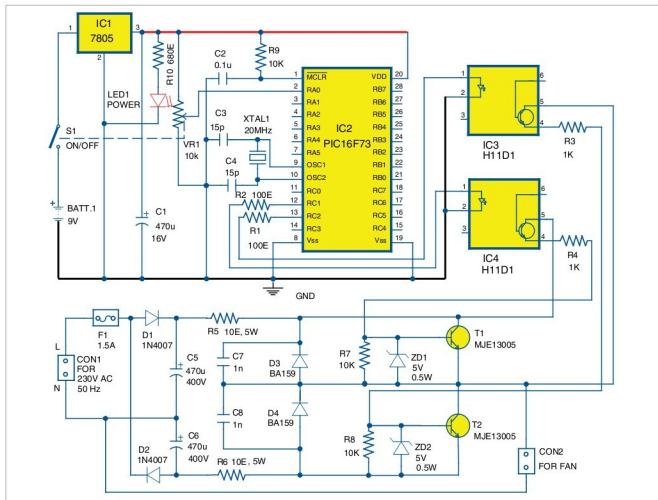


Fig. 6: Circuit diagram of the fan speed increasing regulator

through the base of the switching transistor. Second, its off-state voltage rating must be 300V.

Such optoisolators with 300V rating are available for the phototransistor's V_{CE} value. These include H11D1, SDD400H, SFH618A-5 and HCPL2503, from several manufacturers. Triac drive optoisolator MOC3020 can also be used.

Circuit and working

Having seen how the switches are turned on by PWM1 and PWM2 pulses so as to generate variable-frequency sine wave current into the coil, the block diagram shown in Fig. 5 can be followed easily. Use of an IR remote is optional.

The complete circuit diagram

without IR remote control unit is shown in Fig. 6. It has a PIC16F73 chip programmed with the code written in Assembly language. PWM pulses coming from its pins 12 and 13 have to be checked on a scope. An R-C filter can be used to see the sine waveform as shown in Fig. 2. By varying potentiometer VR1, you can change the frequency of the sine wave.